Cinnamon intake lowers fasting blood glucose: meta-analysis.

Cinnamon, the dry bark and twig of Cinnamomum spp., is a rich botanical source of polyphenolics that has been used for centuries in Chinese medicine and has been shown to affect blood glucose and insulin signaling. Cinnamon's effects on blood glucose have been the subject of many clinical and animal studies; however, the issue of cinnamon intake's effect on fasting blood glucose (FBG) in people with type 2 diabetes and/or prediabetes still remains unclear. A meta-analysis of clinical studies of the effect of cinnamon intake on people with type 2 diabetes and/or prediabetes that included three new clinical trials along with five trials used in previous meta-analyses was done to assess cinnamon's effectiveness in lowering FBG. The eight clinical studies were identified using a literature search (Pub Med and Biosis through May 2010) of randomized, placebo-controlled trials reporting data on cinnamon and/or cinnamon extract and FBG. Comprehensive Meta-Analysis (Biostat Inc., Englewood, NJ, USA) was performed on the identified data for both cinnamon and cinnamon extract intake using a random-effects model that determined the standardized mean difference ([i.e., Change 1(control) - Change 2(cinnamon)] divided by the pooled SD of the post scores). Cinnamon intake, either as whole cinnamon or as cinnamon extract, results in a statistically significant lowering in FBG (-0.49 ± 0.2 mmol/L; $n=8$, $P=.025$) and intake of cinnamon extract only also lowered FBG (-0.48 mmol/L± 0.17; $n=5$, $P=.008$). Thus cinnamon extract and/or cinnamon improves FBG in people with type 2 diabetes or prediabetes.